Assignment 2: Knowledge Representation and Reasoning

FIT5047: Fundamentals of AI — Semester 1, 2026

Juan Pulido
Student ID: 34169202
Faculty of Information Technology, Monash University
Email: jpul0007@student.monash.edu

Question 1: Propositional Logic — Advanced Validity and CNF Transformation

Given the formulas:

- $\phi_1 = ((A \vee (B \to C)) \wedge (\neg C \vee D)) \to ((A \to D) \vee (C \leftrightarrow E))$
- $\phi_2 = (((\neg A \wedge B) \vee (C \to D)) \wedge (\neg E \vee A)) \to ((\neg D \vee F) \leftrightarrow (A \wedge B))$
- $\phi_3 = ((A \leftrightarrow B) \vee (C \to D)) \wedge ((\neg A \vee \neg D) \to (F \wedge \neg G))$

Tasks:

- **(a)** For each formula: Determine whether the clause is valid, satisfiable but not valid, or unsatisfiable. Justify using truth assignments or semantic arguments.
- **(b)** Convert each formula into CNF. Identify Horn clauses where applicable.
- **(c)** For ϕ_2, assume F is always false:
  - (i) What is the effect on satisfiability?
  - (ii) Can you infer anything about A or B? Justify.

Part (a): Validity and Satisfiability

> **Lecture Recap:**
>
> Before we begin, let us establish the logical definitions and rules we will use throughout the solution:
>
> - **Valid:** A formula is true in *all* possible models (truth assignments).
> - **Unsatisfiable:** A formula is false in *all* possible models.
> - **Satisfiable:** A formula is true in *at least one* model.
> - **Satisfiable but not valid:** A formula is true in some models, but false in others.
> - **Horn Clause:** A disjunction of literals (an OR-statement) that contains **at most one positive literal** (e.g., $\neg A \vee \neg B \vee C$).

Strategy to prove validity and satisfiability

- To prove a formula is **satisfiable**, we only need to find one **model** that makes it $True$.
- To prove a formula is **not valid**, we must find a counterexample: a model that makes it $False$.
- Since all three formulas are structured as **implications** ($Antecedent \Rightarrow Consequent$), the only way to make the entire formula $False$ is to make the $Antecent = True$ and the $Consequent = False$.

1. Formula ϕ_1

$\phi_1 = ((A \vee (B \to C)) \wedge (\neg C \vee D)) \to ((A \to D) \vee (C \leftrightarrow E))$

Let $\alpha = ((A \vee (B \to C)) \wedge (\neg C \vee D))$ **the antecedent**
Let $\beta = ((A \to D) \vee (C \leftrightarrow E))$ **the consequent**
So $\phi_1 = \alpha \to \beta$

- **Satisfiability Check:**
  We can easily satisfy the implication by making the $\beta = True$.
  - Let $A = False$ and $D = True$.
  - In the β term $(A \to D)$ becomes $(False \to True) \equiv True$.
  - Since the β is $True$, the entire implication ϕ_1 evaluates to $True$.
  - Thus, ϕ_1 is **satisfiable**.
- **Validity Check:**
  We attempt to find a model where ϕ_1 is $False$ ($\alpha = True$, $\beta = False$).
  - For β to be $False$, both parts of the OR must be $False$:
    - $(A \to D)$ must be $False$ (meaning $A = True, D = False$)
    - $(C \leftrightarrow E)$ must be $False$ (meaning $C \neq E$)
  - For α to be $True$, both parts of the AND must be $True$.
    - Given $D = False$, the second term $(\neg C \vee False)$ means $\neg C$ must be $True$, so $C = False$.
  - Since $C \neq E$ from β and $C = False$ from α, then $E = True$.
  - Looking at the first term of the α: $(A \vee (B \to C))$.
    - Since we know $A = True$, this term is automatically $True$.
  - **Counter-Model:** $\{A = True, B = True, C = False, D = False, E = True\}$.
    - In this model, α is $True$ and β is $False$, yielding $True \to False \equiv False$.

🎯 **Conclusion for ϕ_1:** The formula is **Satisfiable but Not Valid**.

2. Formula ϕ_2

$\phi_2 = (((\neg A \wedge B) \vee (C \to D)) \wedge (\neg E \vee A)) \to ((\neg D \vee F) \leftrightarrow (A \wedge B))$

Let $\alpha = (((\neg A \wedge B) \vee (C \to D)) \wedge (\neg E \vee A))$ the antecedent
Let $\beta = ((\neg D \vee F) \leftrightarrow (A \wedge B))$ the consequent
So $\phi_2 = \alpha \to \beta$

- **Satisfiability Check:**
  We can satisfy ϕ_2 by making the $\alpha = False$.
  - Let $A = False$ and $E = True$.
  - The term $(\neg E \vee A)$ becomes $(False \vee False) \equiv False$.
  - Since the α is $False$, the main implication ϕ_2 evaluates to $True$.
  - Thus, ϕ_2 is **satisfiable**.
- **Validity Check:**
  We attempt to find a counterexample where α is $True$ and β is $False$.
  - Let $D = False$ and $F = False$.
    - The β becomes $(True \vee False) \leftrightarrow (A \wedge B) \equiv True \leftrightarrow (A \wedge B)$.
    - For β to be $False$, $(A \wedge B)$ must be $False$.
    - Let $A = False$, which satisfies $(A \wedge B) = False$.
  - Let $E = False$.
    - The term in α $(\neg E \vee A)$ becomes $(True \vee False) \equiv True$.
  - Now evaluate the remaining part of α: $((\neg A \wedge B) \vee (C \to D))$.
    - Since $A = False$, $\neg A = True$.
    - Let $B = True$, so $(\neg A \wedge B) \equiv True$.
    - Let $C = False$, so $(C \to False) \equiv True$.
    - The α is $True$.
  - **Counter-Model:** $\{A = False, B = True, C = False, D = False, E = False, F = False\}$.
    - In this model, α is $True$ and β is $False$, yielding $True \to False \equiv False$.

🎯 **Conclusion for ϕ_2:** The formula is **Satisfiable but Not Valid**.

3. Formula ϕ_3

$\phi_3 = ((A \leftrightarrow B) \vee (C \to D)) \wedge ((\neg A \vee \neg D) \to (F \wedge \neg G))$

Let $\alpha = ((A \leftrightarrow B) \vee (C \rightarrow D))$ **the left conjunct**
Let $\beta = ((\neg A \vee \neg D) \rightarrow (F \wedge \neg G))$ **the right conjunct**
So $\phi_3 = \alpha \wedge \beta$

- **Satisfiability Check:**
  - Let $A = True, B = True, D = True$.
  - **The** α**:** $(True \leftrightarrow True) \vee \cdots \equiv True$.
  - **The** β**:** The antecedent is $(\neg True \vee \neg True) \equiv False$.
    - An implication with a $False$ antecedent $(False \rightarrow \ldots) \equiv True$.
  - Since both α and β are $True$, ϕ_3 is $True$.
  - Thus, ϕ_3 is **satisfiable**.
- **Validity Check:**
  We can make ϕ_3 $False$ simply by making the $\alpha = False$.
  - Let $A = True, B = False$ (making $A \leftrightarrow B$ $False$).
  - Let $C = True, D = False$ (making $C \rightarrow D$ $False$).
  - The α is $False$, making the entire formula $\phi_3 = False$.

🎯 **Conclusion for** ϕ_3**:** The formula is **Satisfiable but Not Valid**.

Part (b): CNF Transformation and Horn Clauses

Lecture Recap

- **CNF Conversion:** Any logical sentence can be converted into **Conjunctive Normal Form (CNF)**, a conjunction of disjunctions (AND of ORs) by applying four theoretical steps:
  i. **Eliminate biconditionals** $\leftrightarrow$**:** $(A \leftrightarrow B) \equiv (A \rightarrow B) \wedge (B \rightarrow A)$
  ii. **Eliminate implications** $\rightarrow$**:** $(A \rightarrow B) \equiv (\neg A \vee B)$
  iii. **Move** $\neg$ **inwards:** Apply **De Morgan's laws** and **Double Negation**.
  iv. **Distribute** $\vee$ **over** $\wedge$**:** $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$
- **Horn Clause:** A clause (disjunction of literals) containing **at most one positive literal**. These are critical for efficient inference in logical agents.

Transformation of ϕ_1

$\phi_1 = ((A \vee (B \rightarrow C)) \wedge (\neg C \vee D)) \rightarrow ((A \rightarrow D) \vee (C \leftrightarrow E))$

Let $\alpha = ((A \vee (B \rightarrow C)) \wedge (\neg C \vee D))$ **the antecedent**
Let $\beta = ((A \rightarrow D) \vee (C \leftrightarrow E))$ **the consequent**
So $\phi_1 = \alpha \rightarrow \beta$

- **Original**: $\alpha \rightarrow \beta$. Eliminate implication: $\neg\alpha \vee \beta$.
- **Process** α**:** $\alpha = ((A \vee \neg B \vee C) \wedge (\neg C \vee D))$.
  Apply De Morgan's to find $\neg\alpha$:
  $\neg\alpha = (\neg A \wedge B \wedge \neg C) \vee (C \wedge \neg D)$.
- **Process** β**:** $\beta = (\neg A \vee D) \vee (C \leftrightarrow E)$.
  Expand $\leftrightarrow$: $(\neg A \vee D) \vee ((\neg C \vee E) \wedge (\neg E \vee C))$.
  Distribute $\vee$: $\beta = (\neg A \vee D \vee \neg C \vee E) \wedge (\neg A \vee D \vee C \vee \neg E)$.
- **Combine and Simplify:** We distribute $\neg\alpha$ over the two β conjunctions:
  - First conjunct: $\neg\alpha = (\neg A \wedge B \wedge \neg C) \vee (C \wedge \neg D) \vee (\neg A \vee D \vee C \vee \neg E)$.
    Apply rule absorption: Notice $(C \wedge \neg D) \vee D \equiv C \vee D$.
    The expression simplifies to $\alpha = \neg A \vee C \vee D \vee \neg E$.
  - Second conjunct: $\beta = (\neg A \wedge B \wedge \neg C) \vee (C \wedge \neg D) \vee (\neg A \vee D \vee \neg C \vee E)$.
    Apply rule: $(C \wedge \neg D) \vee \neg C \equiv \neg C \vee \neg D$.
    This leaves us with $\beta = D \vee \neg D$, which is a tautology ($\equiv True$)
    So this entire clause drops out.
- **Final CNF for** ϕ_1**:** $(\neg A \vee C \vee D \vee \neg E)$.

- **Horn Clause Check:** It contains two positive literals (C, D), so it is **not** a Horn clause.

Transformation of ϕ_2

$\phi_2 = (((\neg A \wedge B) \vee (C \to D)) \wedge (\neg E \vee A)) \to ((\neg D \vee F) \leftrightarrow (A \wedge B))$

Let $\alpha = (((\neg A \wedge B) \vee (C \to D)) \wedge (\neg E \vee A))$ **the antecedent**
Let $\beta = ((\neg D \vee F) \leftrightarrow (A \wedge B))$ **the consequent**
So $\phi_2 = \alpha \to \beta$

Convert $\phi_2 = \alpha \to \beta \equiv \neg\alpha \vee \beta$

- We again apply $\neg\alpha \vee \beta$.
- **Process $\neg\alpha$:** $\neg(((\neg A \wedge B) \vee (\neg C \vee D)) \wedge (\neg E \vee A))$.
  Pushing negations yields: $\neg\alpha = (A \wedge C \wedge \neg D) \vee (\neg B \wedge C \wedge \neg D) \vee (E \wedge \neg A)$.
- **Process β:** $(D \vee F) \leftrightarrow (A \wedge B)$.
  Converting this generates 5 distinct sub-clauses:
  $(D \vee A) \wedge (D \vee B) \wedge (\neg F \vee A) \wedge (\neg F \vee B) \wedge (\neg A \vee \neg B \vee \neg D \vee F)$.
- **Combine and Simplify:** Distributing $\neg\alpha$ over the β yields 11 distinct clauses:
  - 1: $\neg A \vee \neg B \vee \neg D \vee F$ **(Horn clause)**
  - 2: $\neg B \vee A \vee D \vee E$
  - 3: $A \vee C \vee D \vee E$
  - 4: $B \vee C \vee D \vee E$
  - 5: $\neg A \vee B \vee C \vee D$
  - 6: $A \vee C \vee E \vee \neg F$
  - 7: $A \vee \neg D \vee E \vee \neg F$
  - 8: $B \vee C \vee E \vee \neg F$
  - 9: $\neg A \vee B \vee C \vee \neg F$
  - 10: $B \vee \neg D \vee E \vee \neg F$
  - 11: $\neg A \vee B \vee \neg D \vee \neg F$ **(Horn clause)**
- **Horn Clause Check:** Clauses 1 and 11 contain exactly one positive literal each (F and B, respectively), making them Horn clauses.
  - **Clause 1:** $(\neg A \vee \neg B \vee \neg D \vee F)$ (One positive literal: F)
  - **Clause 11:** $(\neg A \vee B \vee \neg D \vee \neg F)$ (One positive literal: B)

Transformation of ϕ_3

$\phi_3 = ((A \leftrightarrow B) \vee (C \to D)) \wedge ((\neg A \vee \neg D) \to (F \wedge \neg G))$

Let $\alpha = ((A \leftrightarrow B) \vee (C \to D))$ **the left conjunct**
Let $\beta = ((\neg A \vee \neg D) \to (F \wedge \neg G))$ **the right conjunct**
So $\phi_3 = \alpha \wedge \beta$

- Because ϕ_3 is an AND statement between two large expressions, we process them as α and β separately.
- **Process α:** $(A \leftrightarrow B) \vee (C \to D)$.
  - Eliminate equivalents: $((A \vee \neg B) \wedge (\neg A \vee B)) \vee (\neg C \vee D)$.
  - Distribute OR over AND: $(A \vee \neg B \vee \neg C \vee D) \wedge (\neg A \vee B \vee \neg C \vee D)$.
- **Process β:** $(\neg A \vee \neg D) \to (F \wedge \neg G)$.
  - Eliminate implication: $\neg(\neg A \vee \neg D) \vee (F \wedge \neg G)$.
  - Apply De Morgan's: $(A \wedge D) \vee (F \wedge \neg G)$.
  - Distribute OR over AND: $(A \vee F) \wedge (A \vee \neg G) \wedge (D \vee F) \wedge (D \vee \neg G)$.
- **Final CNF for ϕ_3 (combining all 6 clauses):**
  - 1: $A \vee \neg B \vee \neg C \vee D$
  - 2: $\neg A \vee B \vee \neg C \vee D$
  - 3: $A \vee F$
  - 4: $A \vee \neg G$ **(Horn clause)**
  - 5: $D \vee F$
  - 6: $D \vee \neg G$ **(Horn clause)**
- **Horn Clause Check:** Clauses 4 and 6 contain exactly one positive literal (A and D), so they are Horn clauses.

Part (c): Evaluating ϕ_2 when F is always False

> **Lecture Recap:**
>
> - **Vacuous Truth (Implications):** An implication ($P \to Q$) is evaluated as True unless the premise (P) is True and the conclusion (Q) is False. If the premise is False, the implication is **vacuously True**.
> - **Logical Inference (Entailment):** We say $\alpha \models \beta$ if β is True in every **interpretation** where α is True. If satisfying models exist with conflicting truth values for a variable, no definitive entailment exists.

Given $\phi_2 = (((\neg A \wedge B) \vee (C \to D)) \wedge (\neg E \vee A)) \to ((\neg D \vee F) \leftrightarrow (A \wedge B))$.

Let $\alpha = (((\neg A \wedge B) \vee (C \to D)) \wedge (\neg E \vee A))$ **the antecedent**
Let $\beta = ((\neg D \vee F) \leftrightarrow (A \wedge B))$ **the consequent**
So $\phi_2 = \alpha \to \beta$

Assume $F = False$.

(i) What is the effect on satisfiability?

- Substitute $F = False$ into the original implication ϕ_2.
- The $\beta = ((\neg D \vee F) \leftrightarrow (A \wedge B))$.
- With $F = False$,
- The β becomes $((\neg D \vee False) \leftrightarrow (A \wedge B))$, which simplifies directly to $(\neg D \leftrightarrow (A \wedge B))$.
- The modified formula ϕ_2' is: $\alpha \to (\neg D \leftrightarrow (A \wedge B))$.

> 🎯 **Conclusion:** The formula **remains satisfiable**. We can still trivially satisfy it by forcing the α to be $False$ (for example, setting $A = False$ and $E = True$), which makes the entire implication $True$ regardless of the β.

(ii) Can you infer anything about A or B? Justify

- Let us test different satisfying truth assignments (models) for our modified ϕ_2':
  - **Model 1 (A is False, B is True):**
    - Let $A = False, E = True$.
    - The α evaluates to $False$.
    - The implication ϕ_2' holds True.
    - Thus, A can be False.
  - **Model 2 (A is True, B is True):**
    - Let $A = True, B = True, C = False, D = False, E = True$.
    - The α becomes $True$.
    - The β becomes $\neg False \leftrightarrow (True \wedge True) \equiv True \leftrightarrow True \equiv True$.
    - The implication ϕ_2' becomes $True \to True$ holds True.
  - **Model 3 (A is True, B is False):**
    - Let $A = True, B = False, C = False, D = True, E = True$.
    - The α becomes $True$.
    - The β becomes $\neg True \leftrightarrow (True \wedge False) \equiv False \leftrightarrow False \equiv True$.
    - The implication ϕ_2' becomes $True \to True$ holds True.

> 🎯 **Conclusion: No, we cannot unconditionally infer anything about the truth values of A or B.** Because there exist perfectly valid, satisfying models where A and B can independently take either True or False values, the knowledge base is not strict enough to force a specific truth assignment for them.